



NR NW&C Pilot

Pilot project scoping

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Initial Remit

From Mike Evans 24/04/26

- DRN assurance – DRN quality tool which highlights whether PE's are adopting the designer role under CDM, incorrectly commenting on items which are in the submission, improving language for clarity, and flags if a comment is requesting something out of scope (commercial impact).
- CRT development – Our goal would be to use AI to auto populate CRT's (as far as possible) which are tailored to the work and scope described by the DPE.
- Verification and Validation (V&V) Evidence support tool – An ability to automate links demonstrating that success criteria had been met and referencing the appropriate evidence



01 AutoDRN



The Problem

DRN ping-pong

Comments cycle between PE and DPE on tone, clarity and minor wording. Each round absorbs reviewer time without improving the technical position.

Commercial implications

Wording that prescribes method or scope can shift commercial risk onto the PE. Often spotted late, after the DRN has left the contractor.

Duty boundary blurred

Comments directing the designer's method risk PE assuming designer duty under CDM 2015. Boundary calls made case-by-case with no consistent check.

To add

[Add additional pain point or context here]



Volume & Impact

Sizing the opportunity — placeholder figures to be confirmed with DPE/PE input.

[X]

PE / DPE USERS

Number of reviewers across NWC who would use AutomateDRN.

[Y]

DRNS PER 4 WEEKS

DRNs currently generated and reviewed in a typical 4-week cycle.

[Z]%

HIT ONE OF THE 3 PAIN POINTS

Share of comments triggering tone, commercial, or duty-boundary issues.

To add

[Add fourth metric — e.g. reviewer hours per cycle, % rework, or cost impact]



Where AutoDRN plugs in



The Solution — AutomateDRN

Upload DRN → critique each comment against agreed criteria → RAG-rated suggestions with full audit trail.

AutomateX / AutomateDRN

DRN-186891-A01 - review draft

Reviewing

Water Curtain Scheme — Form G submission

007535.00-RBA-FMG-XX-TRP-Z-23001 - Rev B01 - 30 comments by D. Ajiboye, D.S. Owen

TOTAL

30

PASS

14

SOFT FLAG

9

HARD FLAG

7

Comments needing attention

showing 6 of 16 flagged

#1 DA - Form G - Cat 2A

TONE - CLARITY

Original

OS Grid ref should be stated on drawing and not point to the drawing which coincidentally does not have OS Grid reference? There is a reason why NR require it on Form G.

Suggested rewrite - 86% confidence

OS Grid reference should be stated directly on the drawing. The current approach references a drawing which does not itself contain the OS Grid reference. Please update per Form G requirements (NR/L2/CIV/003 §3).

Accept rewriteKeep originalEdit ↗

#8 DA - Drg-30001 - Cat 2A

CDM DESIGNER-ROLE

Original

Setting out information on drawing is inadequate. To set out a corner of a new structure on site, a minimum of two dimensions to different existing known point is necessary.

Risk: prescribes the design method (u201Cminimum two dimensions to different known pointsu201D). PE may be assuming designer role under CDM 2015.

Suggested rewrite - 91% confidence

Setting-out information provided is insufficient to enable site implementation. Please demonstrate that the setting-out information enables unambiguous location of the structure relative to existing assets.

Accept rewriteKeep originalEdit ↗

#11 DA - Drg-30001 - Cat 2A - interface

CROSS-DISCIPLINE

Review dashboard — RAG summary

Rewrite accepted - entry logged

DRN-186891-A01 / log-4

Comment #19 - Drg-30004

Pipework support clarification

BEFORE

How is the pipework on the gantry supported? On the drawing it appears it floating

AFTER

How is the pipework on the gantry supported? On the drawing the pipework appears unsupported. Please clarify.

AUDIT METADATA

Action	Rewrite accepted
Reviewer	Bill Guo (DPE)
Timestamp	27 Apr 2026 - 14:22:08 BST
Classification	Clarity - grammar (soft flag)
Model	AutomateDRN v0.3-alpha
Suggestion confidence	0.92
Original preserved	Yes — recoverable
DRN status	Draft — not yet exported

All changes are provisional until DRN export. Original wording is permanently retained in the audit log regardless of accept/override decisions. PE can revert any single change at any time before export.

Revert changeView full log ↗Next comment ↗

Per-comment audit trail



What This Solves

The DRN review process today — and what AutoDRN changes

Today

- **TIME** DRN ping-pong consumes weeks of DPE / PE rework cycles
- **QUALITY** Comments slip through with tone, clarity or factual issues
- **COMMERCIAL** Drafting risk visible only late — once it has cost implication
- **DUTY** Designer-role exposure in comments goes unflagged (CDM)
- **DPE** Senior engineers absorb low-judgement editing work

With AutoDRN

- **TIME** Single-pass DRNs — most rework eliminated before issue
- **QUALITY** Every comment graded on tone, clarity, factual fit
- **COMMERCIAL** Commercial flags surface at draft, not at dispute
- **DUTY** CDM duty boundary checked — PE stays clear of designer role
- **DPE** DPEs spend their time judging, not editing

Scope

V1

Agreed for v1

- Tone & clarity
- CDM duty boundary check
- Factual — flag only
- Commercial — flag only

V2

RAG pipeline (v2)

- Factual — with project KB
- Commercial implications
- Programme implications
- Standards / ground-truth refs
- Inter-disciplinary checks

+

To add

[Add further criteria, edge cases, or workshop outputs]



02 AutoCRT

Semi-automated CR-T generation with HITL judgement from the DPE



The Problem

Manual CR-T effort

[Add specifics of how much DPE time is absorbed in CR-T drafting and review across a project.]

Inconsistent output

[Add detail on variability between DPEs / projects and downstream rework caused by inconsistent format or content.]

Knowledge fragmentation

[Add detail on dispersed project knowledge — drawings, reports, decisions — not queryable when DPE is drafting CR-T.]

To add

[Add further pain point identified during DPE wireframe workshop.]



Volume & Impact

Sizing the CR-T workload across the DPE community

[X]

DPEs

Engineers producing
CR-Ts across the
portfolio

[Y]

**CR-Ts / 4
weeks**

Drafted, reviewed
and issued in a
typical period

[Z]%

Hit pain points

Share of CR-Ts
requiring rework, late
changes or
knowledge chase

[N]

hrs

Avg DPE time

Cumulative hours per
CR-T — drafting,
reviewing, chasing

Placeholders — confirm with DPE team during wireframe workshop



The Solution

Custom CR-T tooling with project knowledge graph and DPE-led HITL

WORKING CANVAS · SECTION-BY-SECTION ASSEMBLY

Section 4.2 — Project-specific requirements

8 requirements drafted from RRD/F001 · 7 cited · 1 awaiting citation

DOCUMENT TREE

- 1. Introduction ✓
- 2. Engineering Mgmt ✓
- 3. Project Scope ✓
- 4. Requirements ●
- 4.1 Standard ✓
- 4.2 Project-specific ●
- Annex 1 — Deliverables ○
- Annex 2 — PANs ○
- Annex 3 — PCAT ○

7 of 9 sections drafted
2 awaiting input

ACTIVE SECTION · 4.2

Lock section Regenerate

M1/UB1247/4.2/001

CITED 0.91 CONTRACTUAL

The Supplier shall develop a minimum of three (3) viable engineering options for renewal or refurbishment of UB-1247, each demonstrating restoration to RA10 loading capability without enhancement requirements on adjacent assets.

Source: RRD §4 · NR/L2/CIV/003 §3.2 · Acceptance: optioneering report w/ cost-benefit

M1/UB1247/4.2/002

CITED 0.94 CONTRACTUAL

The Supplier shall produce a Form A submission for the recommended option in accordance with NR/L2/CIV/003 Issue 8, accompanied by Form B and any necessary Departures from Standard.

Source: NR/L2/CIV/003 §4 (AIP) · Acceptance: Form A accepted by NW&C Lead DPE

M1/UB1247/4.2/005

UNCITED 0.62 CONTRACTUAL

The Supplier shall produce a Stakeholder Engagement Plan covering the Local Highway Authority, Heritage England, affected freight operators, and adjacent landowners.

△ Inferred from project context (1898 origin → Heritage England) — no source clause found in NR L2/L3 corpus. Supply citation or confirm intent before lock.

M1/UB1247/4.2/008

CITED 0.88 CONTRACTUAL

The Supplier shall produce a Stage 2 cost plan to NR Cost Estimating Standard with confidence levels (P50/P80) clearly stated.

Source: NR Cost Estimating Standard §6.2 · Acceptance: cost plan accepted by NR Cost Manager

+ 4 more requirements

PROJECT CONTEXT

Asset: UB-1247
Stage: 2 (SOS / AIP)
Disciplines: CIV, E&P (Interface)
Driver: RA10 restoration

SOURCES CONSULTED

- RRD/F001 · UB1247 v02
- NR/L2/CIV/003 Issue 8
- NR/L2/RSE/02009 Issue 8
- Closed schemes x 4
- PCAT (filtered to CIV)

SECTION STATUS

7 cited ✓
1 uncited △
8 contractual flag △

AutoCRT does

Data embedding · navigation · retrieval
from project knowledge graph

DPE does (HITL)

Decision · judgement · assurance ·
engineering expertise

*Portfolio-tailored: footbridge · geotech · structural —
same logic, different document conventions*

Section-by-section assembly with cited evidence

What This Solves

The CR-T workflow today — and what AutoCRT changes

Today

- **TIME** DPEs spend days drafting CR-Ts from scratch each project
- **QUALITY** Output varies — inconsistent format and content across DPEs
- **KNOWLEDGE** Project intel scattered — drawings, reports, decisions hunted manually
- **COMMERCIAL** Late-stage CR-T issues drive contract changes and rework
- **DPE** Senior engineers buried in document assembly, not judgement

With AutoCRT

- **TIME** CR-T drafted in hours, not days — DPE reviews and refines
- **QUALITY** Consistent structure and language across DPEs and projects
- **KNOWLEDGE** Queryable knowledge graph — every clause traceable to source
- **COMMERCIAL** Contract clarity at first issue — fewer changes, fewer disputes
- **DPE** Engineering judgement front-and-centre; assembly automated

CDE · Where AutoCRT plugs in



OPEN QUESTION

Which CDE for the project knowledge layer? SharePoint · eB · ProjectWise · CEMAR — to confirm with project



Data & Scope

What we need from the project to deliver · What AutoCRT is not

Data we need

- Past CR-Ts — portfolio-tagged where applicable (footbridge / geotech / ...)
- CR-T guidance notes and any internal authoring conventions
- Source documents to feed the project knowledge graph (drawings, reports, decisions)
- DPE wireframe of the existing CR-T workflow

What AutoCRT is not

- Not redefining how a CR-T should be generated — we replicate the agreed workflow
- Not training material or DPE guidance — it is an automation tool
- Not automating PCAT — deliverables remain a manual DPE decision

03

AutomateV&V

Golden-thread traceability from RRD through to operational acceptance



The Problem

Broken golden thread

Requirements traced manually — RRD → CRT → design → as-built → EIS links lost or unevidenced between stages.

Audit-pack scramble

Evidence assembled retrospectively at gate reviews — weeks of DPE time chasing documents that already exist.

Semantic drift

Wording shifts between RRD and contract — obligations subtly change without anyone noticing until dispute or claim.

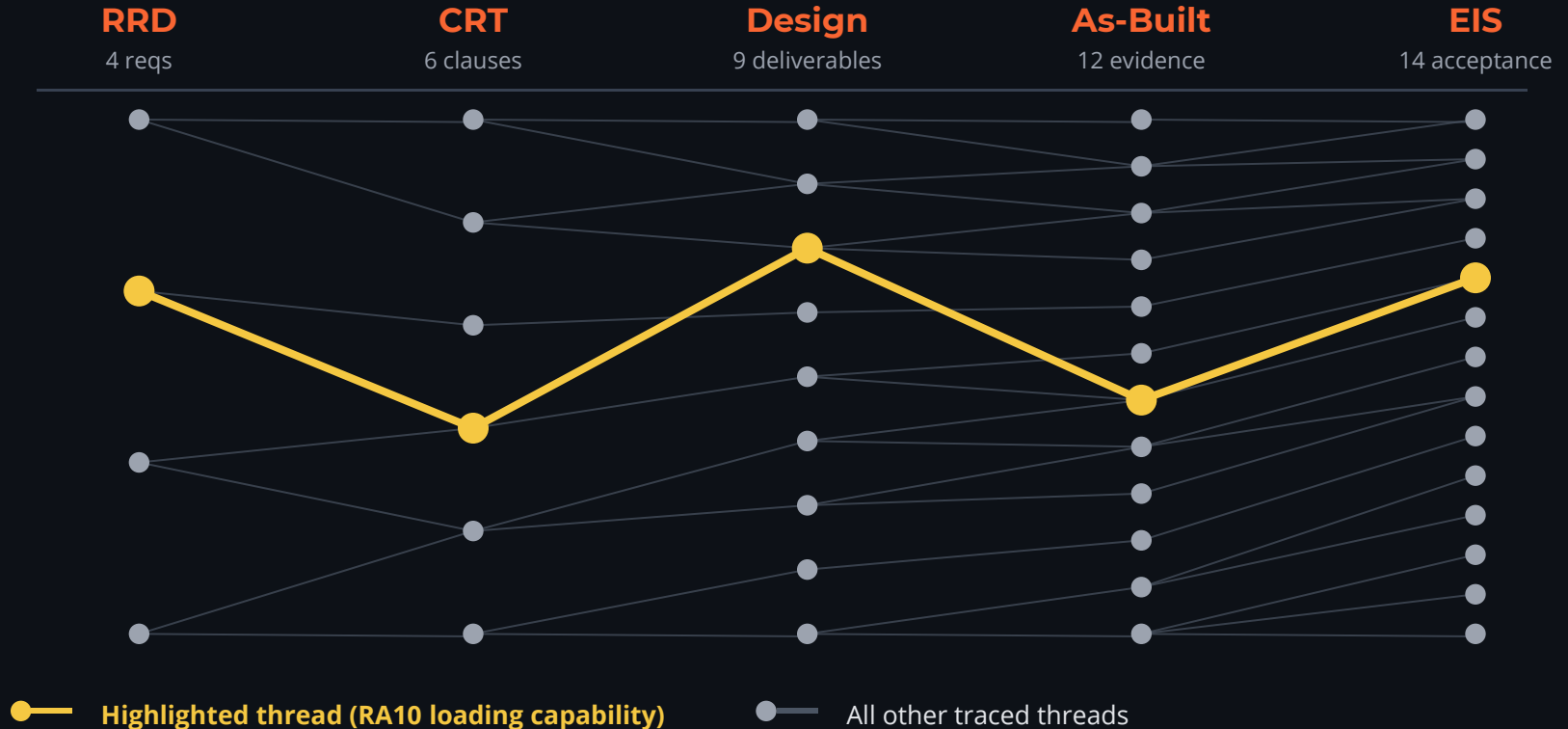
To add

[Add further pain point identified during stakeholder review.]



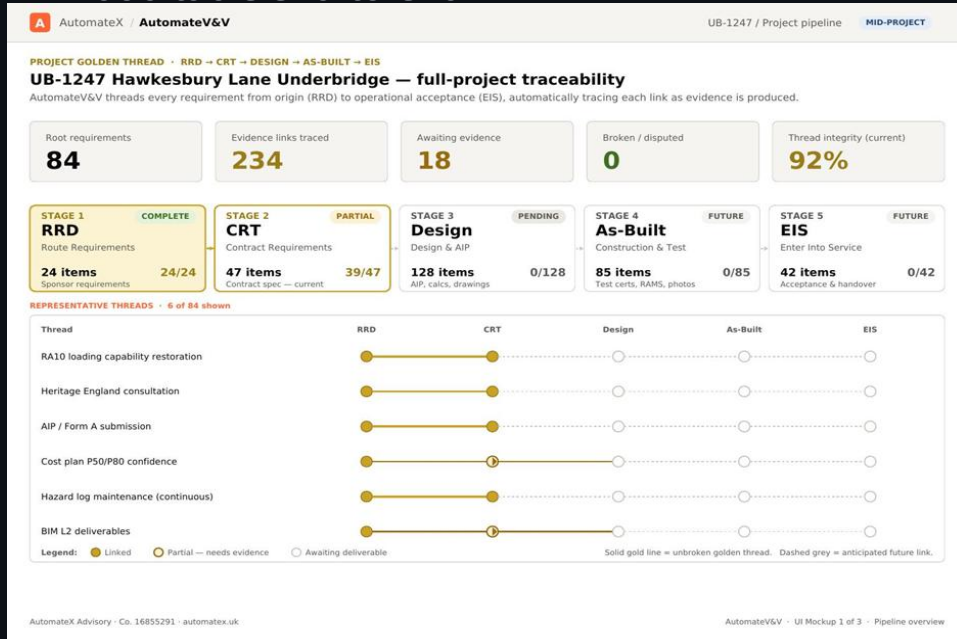
The Golden Thread

Few RRD requirements fan out into many EIS evidence items — every link traceable

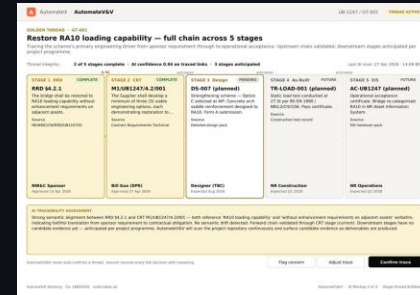


The Solution

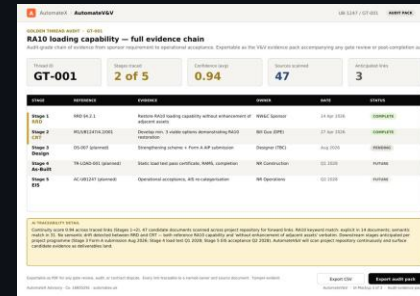
Continuous traceability from sponsor requirement to operational acceptance — auditable end-to-end



Project pipeline · golden-thread integrity at a glance



Single thread · 5 stages assessed



Audit pack · exportable evidence chain

What This Solves

Traceability today — and what AutomateV&V changes

Today

- **TIME** Audit packs assembled retrospectively — weeks of evidence chase at every gate
- **QUALITY** Broken golden thread — evidence links lost between stages
- **RISK** Semantic drift between RRD and contract goes unnoticed
- **COMMERCIAL** Disputes hard to defend without complete evidence chain
- **DPE** Senior engineers absorbed in document chase, not oversight

With AutomateV&V

- **TIME** Audit-ready continuously — evidence pack exportable on demand
- **QUALITY** Unbroken golden thread from sponsor requirement to acceptance
- **RISK** Semantic drift detected at point of issue — every link assured
- **COMMERCIAL** Tamper-evident chain stands up to audit, claim or dispute
- **DPE** Engineering oversight on the chain itself, not the chase

CDE · Where AutomateV&V plugs in



OPEN QUESTION

Which CDE for the project repository scan? SharePoint · eB · ProjectWise · CEMAR — to confirm with project



Data & Scope

What we need from the project to deliver · What AutomateV&V is not

Data we need

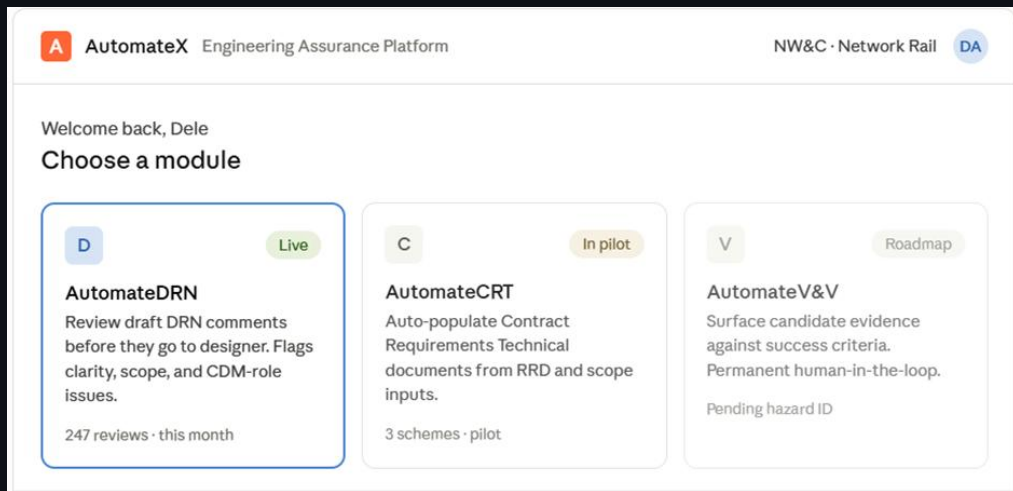
- Read access to the project repository (drawings, RRDs, CRTs, design packs, test certs, EIS handover)
- Past gate-review packs and audit packs as exemplars
- Stage / discipline taxonomy used on the project

What AutomateV&V is not

- Not auto-confirming threads — every link decision recorded by a named assessor
- Not replacing the assessor or the gate review process
- Not building the CDE — we read from the existing repository
- Not generating new evidence — we trace evidence as it is produced

One Platform

All three modules integrate into the AutomateX Engineering Assurance Platform — custom-built for NW&C



Module dashboard

AutoDRN · Live

DRN review with tone, clarity, CDM duty checks

AutoCRT · In pilot

CR-T generation with project knowledge graph

AutomateV&V · Roadmap

Golden-thread traceability, gate-ready evidence

Single sign-on · shared project knowledge · unified audit trail · NW&C-branded

THE BIG PRIZE

An AI-enabled DPE workforce.

Freed up to do the work that matters

Every DPE and PE on the NW&C programme spends their hours on judgement, decisions and risk — not on repetitive document chase, format-policing or evidence assembly. AutomateX solutions help absorb the low-value tasks so the team operates at the level they are paid to operate.

Repetitive tasks absorbed by
the platform

Engineers focused on
decisions and risk

Capability built into the
system, not the room